ICNAAM 2026
24th International Conference of Numerical Analysis and Applied Mathematics

Trapped Modes and BICs in Perturbed Waveguides: A Numerical Validation

Producto derivado del proyecto IMP CIAS 4084 vigencia 2025 I_2026_II

M.I. Romero Rodríguez · D.F. Zárate Bello
K.S. Sepúlveda García · H. Machuca Balaguera

Universidad Militar Nueva Granada · Bogotá, Colombia

Universidad Militar Nueva Granada

ApplyMath

Mathematical formulation

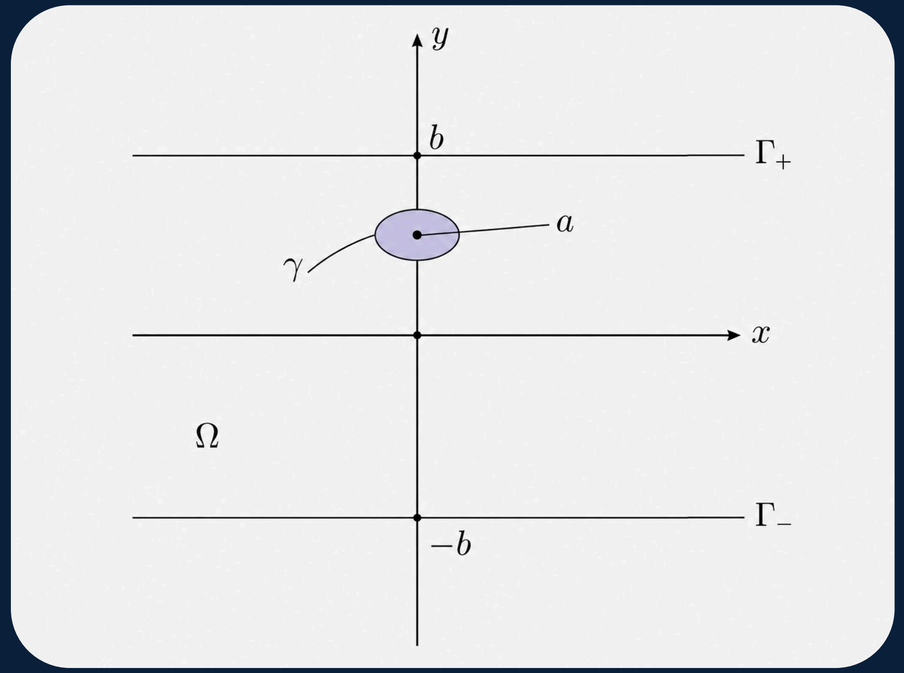
A localized Helmholtz eigenvalue problem

$$
\begin{aligned}
-\Delta u &= k^2 u, && \text{in } \Omega,\\
u &= 0, && \text{on } \Gamma_{\pm},\\
\frac{\partial u}{\partial n} &= 0, && \text{on } \gamma,\\
u(x,y) &\to 0, && \text{as } |x| \to \infty.
\end{aligned}
$$

$$
\Omega = \{(x,y) : |y| < b\} \setminus \overline{\gamma},
$$

$$
\gamma(t) = (\varepsilon X(t),\, a + \varepsilon Y(t)), \quad t \in [0, 2\pi) \quad k^2 \text{ is a spectral parameter}
$$

k it is a spectral parameter



$a = vertical\ displacement\ of\ the\ obstacle's\ center,$
$\varepsilon = obstacle\ scale\ parameter$

Why localized states in an infinite guide?

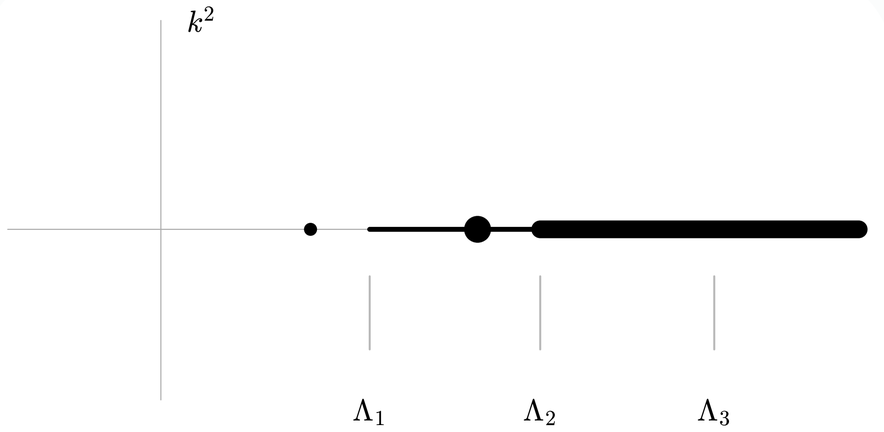
A small, localized perturbation can generate a mode that remains spatially confined.

Discrete trapped mode

Below the propagation threshold.

Bound state in the continuum (BIC)

Inside the continuous spectrum; more delicate numerically.



Continuous spectrum, $\Lambda_n = \pi^2 n^2 / 4b^2$

Does BEM recovertheasymptotic κ=kb neartherelevantthreshold?

Spectral behavior

The distinction is not just semantic: it changes what must be validated numerically.

Small-obstacle BIC: symmetry is required

DISCRETE REGION

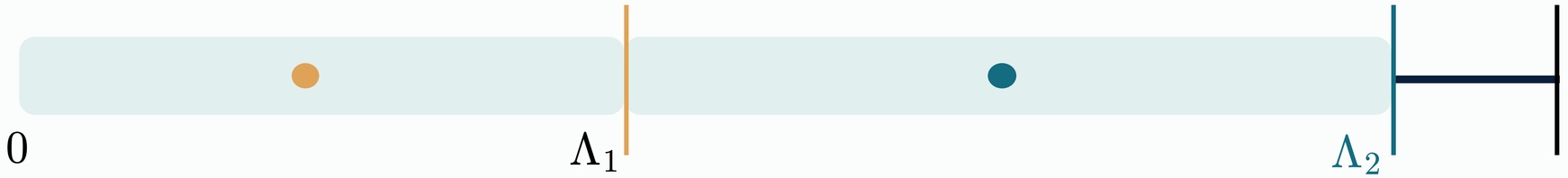
$$k^2 < \Lambda_1$$

No propagating channels are open.

FIRST PROPAGATION INTERVAL

$$\Lambda_1 < k^2 < \Lambda_2$$

First channel open — localization requires cancellation.



cutoff values: $\Lambda_n = \left(\frac{n\pi}{2b}\right)^2, n = 1, 2, 3, \ldots$

$$\sigma = \varepsilon^2 \frac{\pi^2}{4b^3}\left(\pi\mu \sin^2\frac{\alpha}{2} - \frac{1}{2}S\cos^2\frac{\alpha}{2}\right) + \mathcal{O}(\varepsilon^3 \ln \varepsilon)$$

Eq. (2.5) in [10]

$$\sigma = \varepsilon^2 \frac{\pi^3}{b^3}\mu + \mathcal{O}(\varepsilon^3 \ln \varepsilon)$$

Eq. (2.6) in [10]

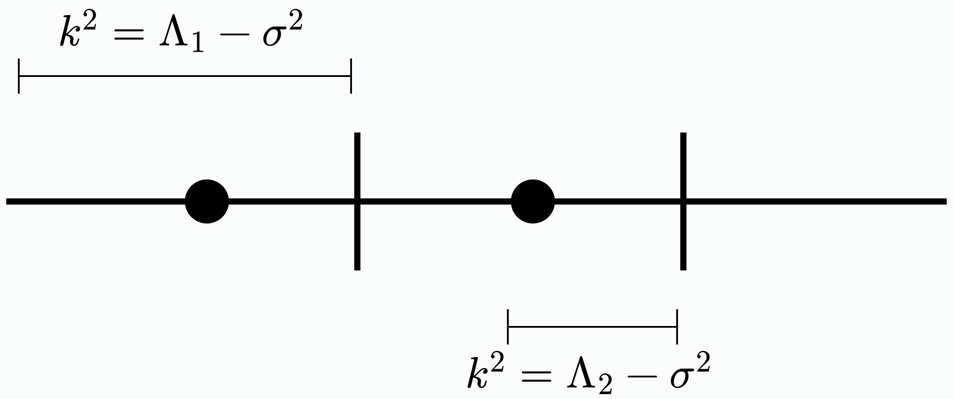
Asymptotic predictions

Objective: assess the agreement of BEM with the leading asymptotic prediction as ε→0.

VALID REGIME

$$\varepsilon \ll 1$$

DISCRETE MODE



BIC NEAR THE SECOND THRESHOLD

Why Boundary Element Method?

The key numerical advantage is structural: the infinite guide is retained.

DOMAIN METHOD

Discretizes a 2D region

Requires longitudinal truncation
PML or DtN treatment

BEM

Discretizes only the obstacle boundary

Waveguide Green function represents
the unbounded domain directly

no artificial truncation

Boundary Element Method formulation

The waveguide Green functions

$$u(P) = \int_{\gamma} u(q)\, \frac{\partial G_s(P,q)}{\partial n_q}\, ds_q.$$

$$G_{\mathrm{s}}(P,q) \sim \sum_{n=0}^{j_s} \alpha_n^{\pm}(q) \cos\left(\frac{n\pi y}{d}\right) e^{\pm ikxt_n}$$

$$\frac{\partial}{\partial n_q} G_{\mathrm{s}}(P,q) \sim \sum_{n=0}^{j_s} \beta_n^{\pm}(q) \cos\left(\frac{n\pi y}{d}\right) e^{\pm ikxt_n}$$

Boundary Element Method formulation

The waveguide Green function satisfies the wall conditions, so only the obstacle boundary (gamma) is discretized. [11]

Waveguide Green Function & Regularization

$$G_s(P,q) \triangleq G_D^{2b}(X,\bar{y},\bar{\eta}) = G_0^{4b}(X,\bar{y}-\bar{\eta}) - G_0^{4b}(X,\bar{y}+\bar{\eta})$$

Eq. (4.5) in [11]

$$X = x-\xi, \qquad \bar{y} = y+b, \qquad \bar{\eta} = \eta+b$$

$$G_{s,\mathrm{reg}}(P,q) = G_s(P,q) - \frac{1}{4}\,Y_0(k|P-q|)$$

$$G_0^d(X,Y) = -\frac{i}{4}\left[H_0^{(1)}(kr) + \sum_{\ell=0}^{\infty} \epsilon_\ell\, S_\ell\, J_\ell(kr)\cos\left[\ell\left(\frac{\pi}{2}-\theta\right)\right]\right]$$

Eq. (2.46) in [11]

$$r = \sqrt{X^2+Y^2}, \qquad \theta = \operatorname{atan2}(Y,X), \qquad \epsilon_0 = 1, \quad \epsilon_\ell = 2 \;\; (\ell > 0)$$

Boundary Element Method formulation

The waveguide Green function satisfies the wall conditions, so only the obstacle boundary (gamma) is discretized. [11]

1 BOUNDARY-INTEGRAL EQUATION

$$u(P) = \int_\gamma \left[u(q) \frac{\partial G_s(P,q)}{\partial n_q} - G_s(P,q) \frac{\partial u(q)}{\partial n_q} \right] ds_q$$

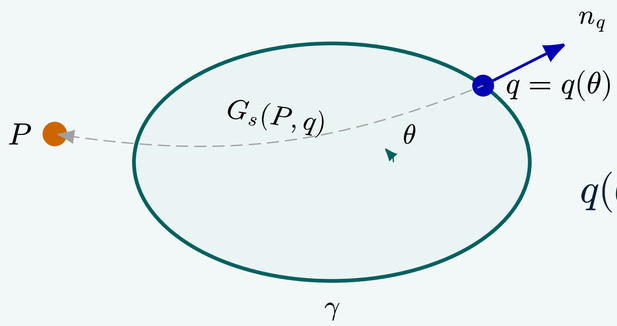
Rigid obstacle:

$$u(P) = \int_\gamma u(q) \frac{\partial G_s(P,q)}{\partial n_q} ds_q$$

Taking $P \to p \in \gamma$ gives the jump relation

$$\frac{1}{2} u(p) = \int_\gamma u(q) \frac{\partial G_s(p,q)}{\partial n_q} ds_q$$

2 GEOMETRY & GREEN KERNEL



$$q(\theta) = \big(X(\theta), a + Y(\theta)\big)$$
$$0 \le \theta < 2\pi$$

$$ds = w(\theta)\, d\theta,$$
$$w(\theta) = \sqrt{X'(\theta)^2 + Y'(\theta)^2}$$

$$K^{(w)}(\psi,\theta) = w(\theta) \frac{\partial G_s}{\partial n_q}$$

3 FULL-CONTOUR MIDPOINT COLLOCATION

$$\theta_i = \left(i + \frac{1}{2}\right) \frac{2\pi}{M}, \qquad i = 0, \dots, M-1$$

$$\frac{1}{2} u_i = \frac{2\pi}{M} \sum_{j=1}^{M} K_{ij}^{(w)}(k) u_j$$

$$A(k)\mathbf{u} = \mathbf{0}$$
$$A(k) = I - \frac{4\pi}{M} K^{(w)}(k)$$

Diagonal $i = j$: use $G_{s,\mathrm{reg}}$ plus the analytic curvature term; off-diagonal term it is evaluated by perturbing the source point q using centred differences, with $h = 10^{-6}$

Discrete trapped mode: validation workflow

Theorem 2.1 - local search plus uniqueness screening in $[0, \Lambda_1)$

SEARCH WINDOW

1. Asymptotic predictor

Compute kb_{asym} and σ_{asym}

Form small obstacle formula

→

2. Physical band

Restrict to the discrete intervalA

$0 < k^2 < \Lambda_1$

→

3. Whole band screen

Sample $\sigma_{\min}(A(kb))$ across the band

Detect local minima

→

4. Expected-mode refinement

Search near kb_{asym}

$[kb_{asym} - \delta,\ kb_{asym} + \delta]$

bound minimization of $\sigma_{min}(A(kb))$

↓

5. Mesh refinement

Repeat with $M = 16, 24, 32, 40, 48$

At the minimizer $(Kb)^*$

Reassemble $A(Kb)^*)$

Compute the smallest singular value

Extract the right singular vector

$\sigma_{min}(A((kb)^*)) \leq 10^{-6}$

Acceptance tests

Exactly one resolved minimum in $0 < k^2 < \Lambda_1$

The minimum is interior and sufficiently deep

kb and σ are stable under mesh refinement

Compare k_{BEM} and σ_{BEM} with asymptotic predictions

Embedded trapped mode (BIC): numerical validation workflow

Theorem 2.3 - spectral search in $\Lambda_1 < k^2 < \Lambda_2$ plus a non-radiating check

SEARCH WINDOW

1. Asymptotic predictor

Compute kb_{asym} and σ_{asym}

from the BIC asymptotic formula

2. Symmetry setup

x-symmetry: $a = 0$

y-symmetry: tune a near a_{asym}

build the obstacle geometry

3. Physical band

Restrict to the discrete intervalA

$0 < k^2 < \Lambda_1$

4. Expected-mode refinement

Search near kb_{asym}

$[kb_{asym} - \delta,\ kb_{asym} + \delta]$

bound minimization of $\sigma_{min}(A(kb))$

5. Mesh refinement

Repeat with

$M = 16, 24, 32, 40, 48$

At the minimizer $(Kb)^*$

Reassemble $A(Kb)^*)$

Compute$\sigma_{min}$and the right singular vector

Reconstruct the field u

$\sigma_{min}(A((kb)^*)) \leq 10^{-6}$

BIC-only propagating-mode test

Evaluate u at $x = \pm 2b$

Project onto the first open Dirichlet mode

$$R_{left},\ R_{right} = \frac{\|\Pi_{prop}\, u\|}{\|u\|}$$

Accept only if $R_{left} \ll 1$ and $R_{right} \ll 1$

Agreement with asymptotic

Relative wavenumber error between BEM and asymptotic prediction.

ε	Discrete	BIC X	BIC Y
0.05	3.85×10^{-7}	1.38×10^{-5}	1.39×10^{-5}
0.07	3.26×10^{-6}	1.19×10^{-4}	1.20×10^{-4}
0.09	1.59×10^{-5}	5.68×10^{-4}	5.73×10^{-4}
0.11	5.50×10^{-5}	1.92×10^{-3}	1.93×10^{-3}

$$E_k = \frac{|kb_{\mathrm{BEM}} - kb_{\mathrm{asym}}|}{|kb_{\mathrm{asym}}|}$$

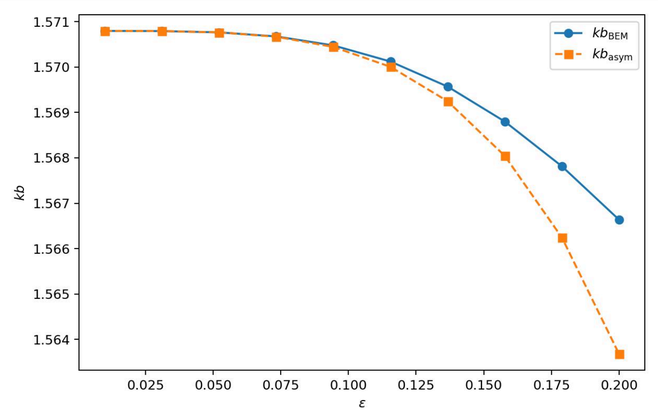
3 FULL-CONTOUR MIDPOINT COLLOCATION

$b = 1.0$ in all cases
Discrete: $a = 0.6$
BIC-X: $a = 0$
BIC-Y: tuned $a = \varepsilon a_1 + O\big(\varepsilon^2 \ln \varepsilon\big)$,
$a_1 = -\beta/12 + O(\beta^2)$ [10]

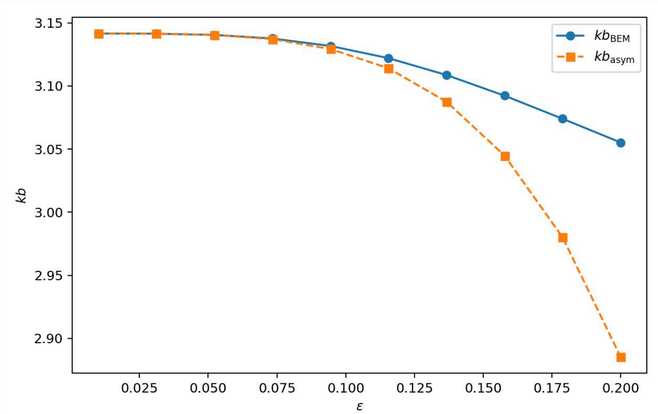
Agreement as obstacle size increases

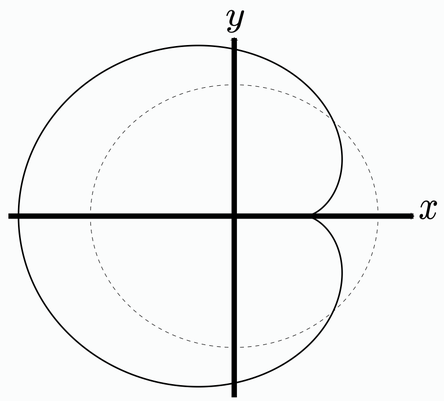
At small ε the curves nearly coincide; separation grows consistently with a small-obstacle expansion.

CIRCLE DISCRETE EINGENVALUE



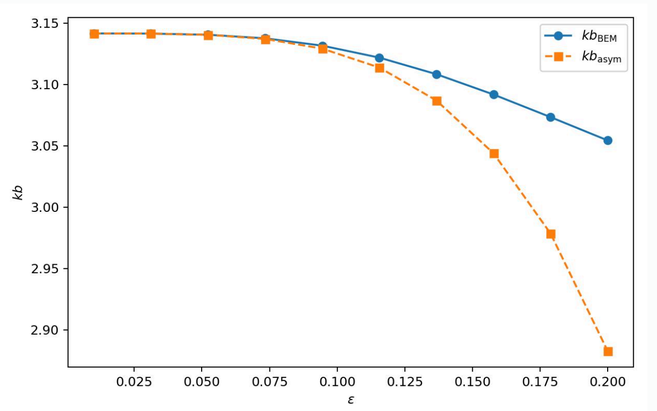
x-SYMMETRIC EMBEDDING TRAPPED MODES

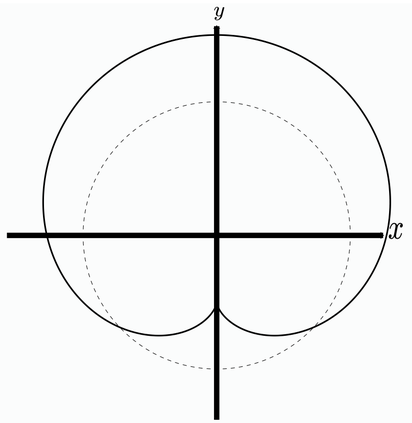




$X(-t) = X(t),\ Y(-t) = -Y(t)$

y-SYMMETRIC EMBEDDING TRAPPED MODES





$X(-t) = X(t),\ Y(-t) = -Y(t)$

Failure of the screening criterion does not prove that a BIC ceases to exist.

Mesh refinement & BIC validation

Representative wavenumbers are stable as the obstacle boundary is refined.

ε	Δkb	Δkb	Δkb
0.05	3.6×10^{-11}	1.37×10^{-8}	1.34×10^{-8}
0.07	8.3×10^{-11}	1.18×10^{-7}	1.08×10^{-7}
0.09	2.39×10^{-9}	3.99×10^{-7}	3.71×10^{-7}
0.11	1.12×10^{-8}	1.01×10^{-6}	1.11×10^{-6}

Mesh refinement produces only negligible changes in k_b, confirming numerical stability for the discrete mode and both BIC configurations.

Conclusions

A quantitative, reproducible bridge between numerical computation and asymptotic theory.

01

Infinite domain, directly represented

The waveguide Green function avoids artificial longitudinal truncation.

02

Quantitative validation at small obstacle size

For representative cases, BEM and asymptotic wavenumbers differ by less than 0.017%.

03

BIC screening beyond singular values

Mesh refinement is complemented by cancellation of the first propagating component.

04

Toward an independent spectral validation

Beyn's method enables an independent spectral search. For Theorem 2.1, it reproduces the BEM eigenvalues to about 10e-8 in kb and supports uniqueness. Extension to the BIC cases is ongoing.

FUTURE WORK

Explore whether larger obstacles produce additional branches of embedded trapped modes.

References

[1] Evans, D. V., Levitin, M., Vassiliev, D. Existence theorems for trapped modes. Journal of Fluid Mechanics 261, 21–31 (1994).

[2] Linton, C. M., McIver, P. Handbook of Mathematical Techniques for Wave/Structure Interactions. Chapman & Hall/CRC, Boca Raton (2001).

[3] McIver, P., Linton, C. M., McIver, M. Construction of trapped modes for wave guides and diffraction gratings. Proceedings of the Royal Society A 454, 2593–2616 (1998).

[4] Hein, S., Koch, W. Acoustic resonances and trapped modes in pipes and tunnels. Journal of Fluid Mechanics 605, 401–428 (2008).

[5] Exner, P., Šeba, P. Bound states in curved quantum waveguides. Journal of Mathematical Physics 30, 2574–2580 (1989).

[6] Duclos, P., Exner, P. Curvature-induced bound states in quantum waveguides in two and three dimensions. Reviews in Mathematical Physics 7, 73–102 (1995).

[7] von Neumann, J., Wigner, E. Über merkwürdige diskrete Eigenwerte. Physikalische Zeitschrift 30, 465–467 (1929).

[8] Hsu, C. W., Zhen, B., Stone, A. D., Joannopoulos, J. D., Soljačić, M. Bound states in the continuum. Nature Reviews Materials 1, 16048 (2016). doi:10.1038/natrevmats.2016.48.

[9] Koshelev, K. L., Sadrieva, Z. F., Shcherbakov, A. A., Kivshar, Yu. S., Bogdanov, A. A. Bound states in the continuum in photonic structures. Physics-Uspekhi 66, 494–517 (2023). doi:10.3367/UFNe.2021.12.039120.

[10] Zhevandrov, P., Merzon, A., Romero Rodríguez, M. I., De la Paz Méndez, J. E. Discrete and embedded trapped modes in a plane quantum waveguide with a small obstacle: exact solutions. Acta Applicandae Mathematicae 196, 8 (2025). doi:10.1007/s10440-025-00720-2.

[11] Linton, C. M. The Green's function for the two-dimensional Helmholtz equation in periodic domains. Journal of Engineering Mathematics 33, 377–401 (1998). doi:10.1023/A:1004377501747.

[12] Chen, J. T., Hong, H. K. Review of dual boundary element methods with emphasis on hypersingular integrals and divergent series. Applied Mechanics Reviews 52, 17–33 (1999).

[13] Tai, C. H., Chen, J. T., Kao, I. L. A new SVD-updating technique for evaluating true and spurious eigenvalues in BEM. Engineering Analysis with Boundary Elements 27, 167–174 (2003).